

PROBLEM STATEMENT #24 | Smart Cities, Transport & Logistics**Public Bus Live Tracking & Crowding Estimator****1. Problem Context & Overview**

A municipal transit intelligence platform ingesting GPS feeds from city buses, estimating arrival times (ETA) at downstream stops, and computing passenger crowding levels from ticketing sensors.

Target Stakeholders / Actors: *Commuter, Fleet Controller*

2. Sample Functional Requirement (FR) Guideline**Sample Requirement: FR-001 [Priority: High]**

Description: The system shall ingest bus GPS coordinates every 5 seconds and recalculate arrival ETAs for all upcoming route stops.

Sample Acceptance Criteria: Pass: ETA updates based on current traffic speed and distance; Fail: ETA increases while bus approaches stop.

3. Sample Non-Functional Requirement (NFR) Guideline**Sample Requirement: NFR-001 [Type: Performance & Security]**

Description: The tracking engine shall handle telemetry streams from up to 1,000 active transit vehicles simultaneously.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]**• Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

• UML Use-Case Diagram:

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

• Use-Case Flow Specification:

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.